

Gallbladder Disease Classification

Using Residual Attention CNN Model

Assignment Report

Name	Sakib Rayhan Yeasin
Student ID	1000128462

1. Introduction

This project classifies gallbladder conditions from ultrasound images using a custom Residual Attention Convolutional Neural Network (CNN) developed in PyTorch. The dataset contains 10,692 images from 9 classes. The objective is to predict the correct gallbladder condition from an input ultrasound image.

2. CNN Architecture and Methodology

2.1 Data Preparation

The dataset was divided into training, validation, and test sets. Duplicate and visually similar images were checked before splitting. Similar images were kept in the same split to reduce the risk of data leakage.

Training	Validation	Test
8,554 images	1,067 images	1,071 images

All images were converted to RGB, resized to 100 x 100 pixels, converted to PyTorch tensors, and scaled to values between 0 and 1.

2.2 Model Architecture

The model is a custom Residual Attention CNN with about 2.7 million trainable parameters. Residual connections help information flow through the network, while channel attention helps the model focus on useful feature channels.

Stage	Main Operation
Input / first layer	RGB 3 x 100 x 100 -> 3 x 3 convolution, 32 filters
Residual attention	Three blocks with 64, 128, and 256 filters
Feature summary	512-channel convolution + global average and max pooling
Classifier	Fully connected layers: 1024 -> 256 -> 128 -> 9

The network uses SiLU activation, Batch Normalization, and Dropout. The final layer produces scores for the 9 classes.

2.3 Training Methodology

The model was trained for 20 epochs with a batch size of 32. Adamax was used as the optimizer and Cross-Entropy Loss was used for classification. L2 regularization was added to the dense layers, and the learning rate was gradually reduced during training.

Setting	Value	Setting	Value
Epochs	20	Batch size	32
Optimizer	Adamax	Initial LR	0.001
Loss	Cross-Entropy	L2	0.0001

3. Training and Evaluation Results

The model learned quickly during training and achieved high accuracy on the validation and test sets. The final test accuracy was 97.85%, and the macro F1 score was 97.93%.

Metric	Result
Training Accuracy	99.99%
Validation Accuracy	98.03%
Test Accuracy	97.85%
Macro Precision	97.89%
Macro Recall	98.02%
Macro F1 Score	97.93%

Macro AUC: 99.97%

On the test set, the model correctly classified 1,048 of 1,071 images, with 23 incorrect predictions. Carcinoma had the lowest recall at about 92.50%, while the other classes also showed strong performance.

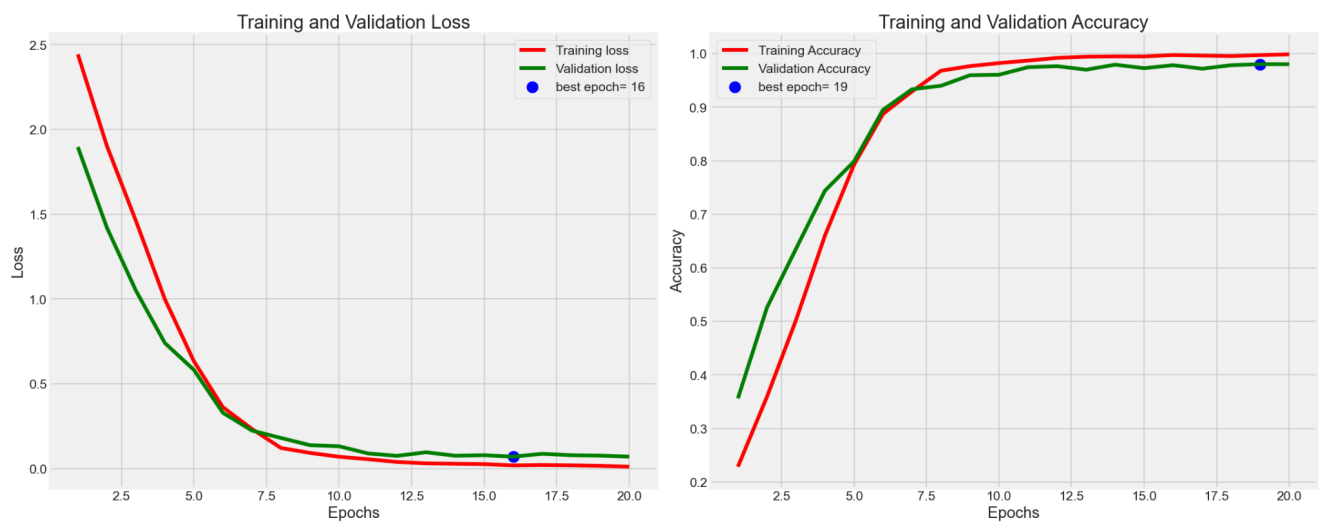


Figure 1. Training and validation accuracy/loss

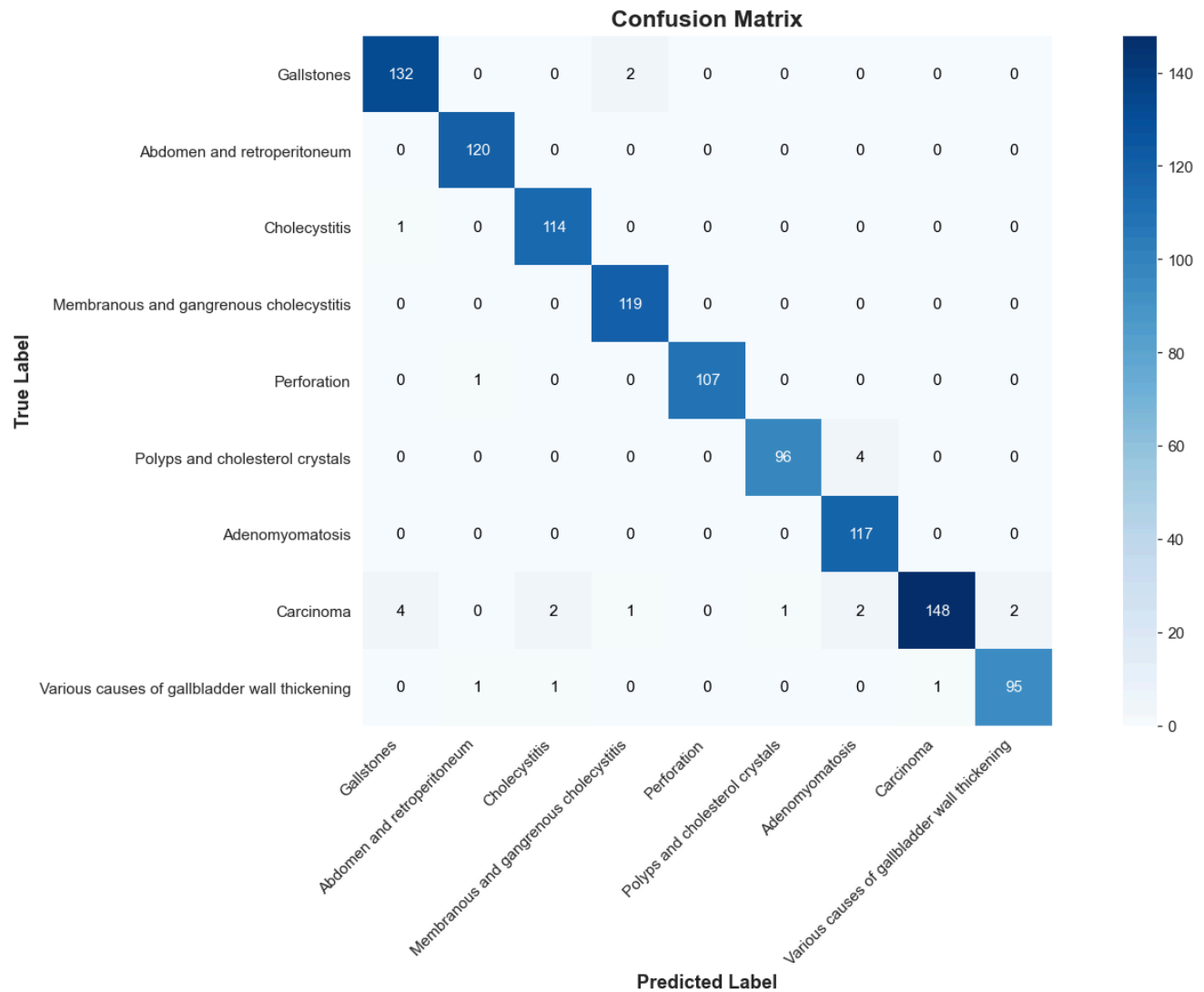


Figure 2. Confusion matrix

4. Evaluation Figures

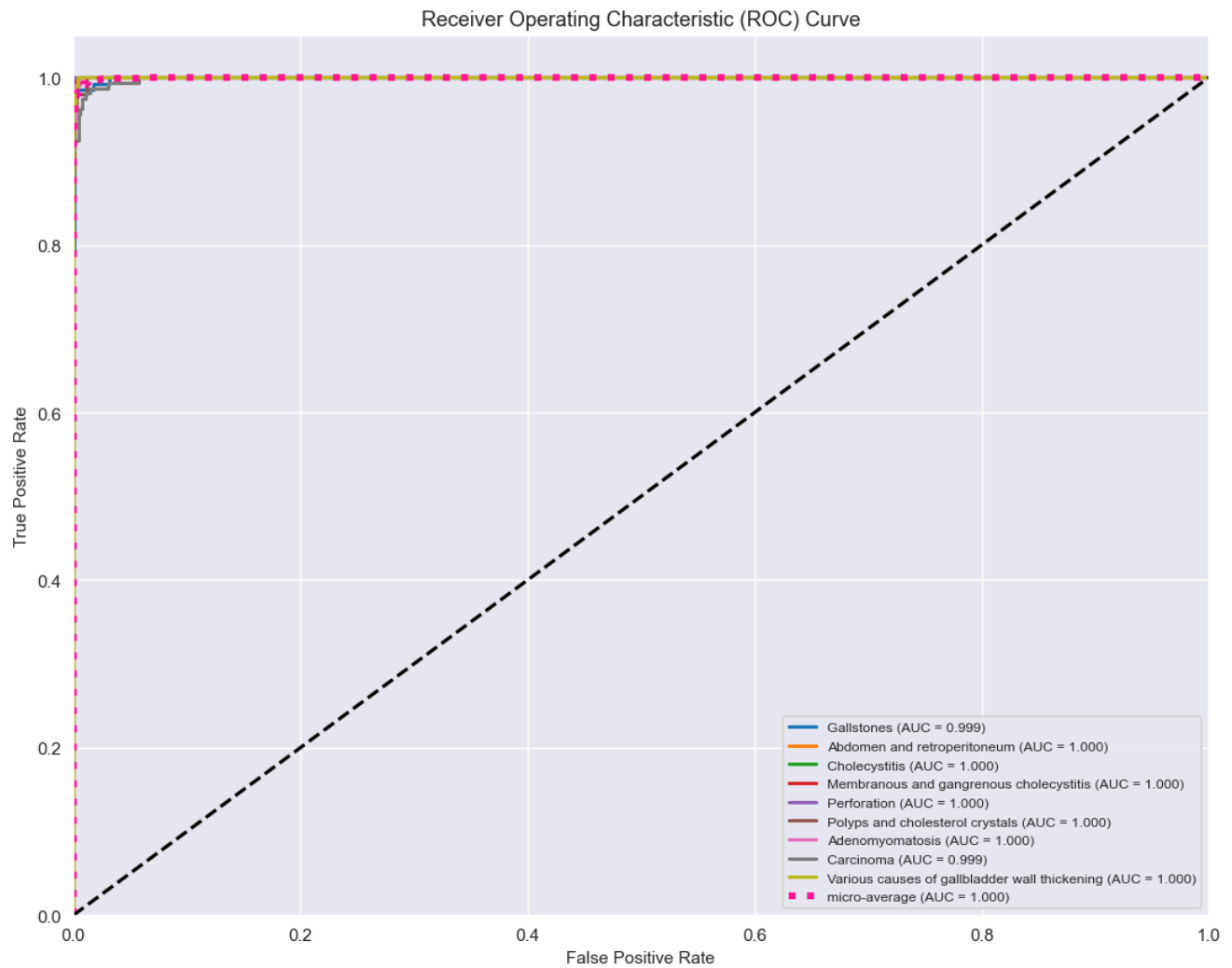


Figure 3. ROC curve

The ROC results show very strong class separation. The reported macro AUC is 99.97%.

5. Sample Output and Prediction Images

The following space is reserved for sample predictions generated by the trained model. Add prediction images from the notebook here.

Random test-set predictions

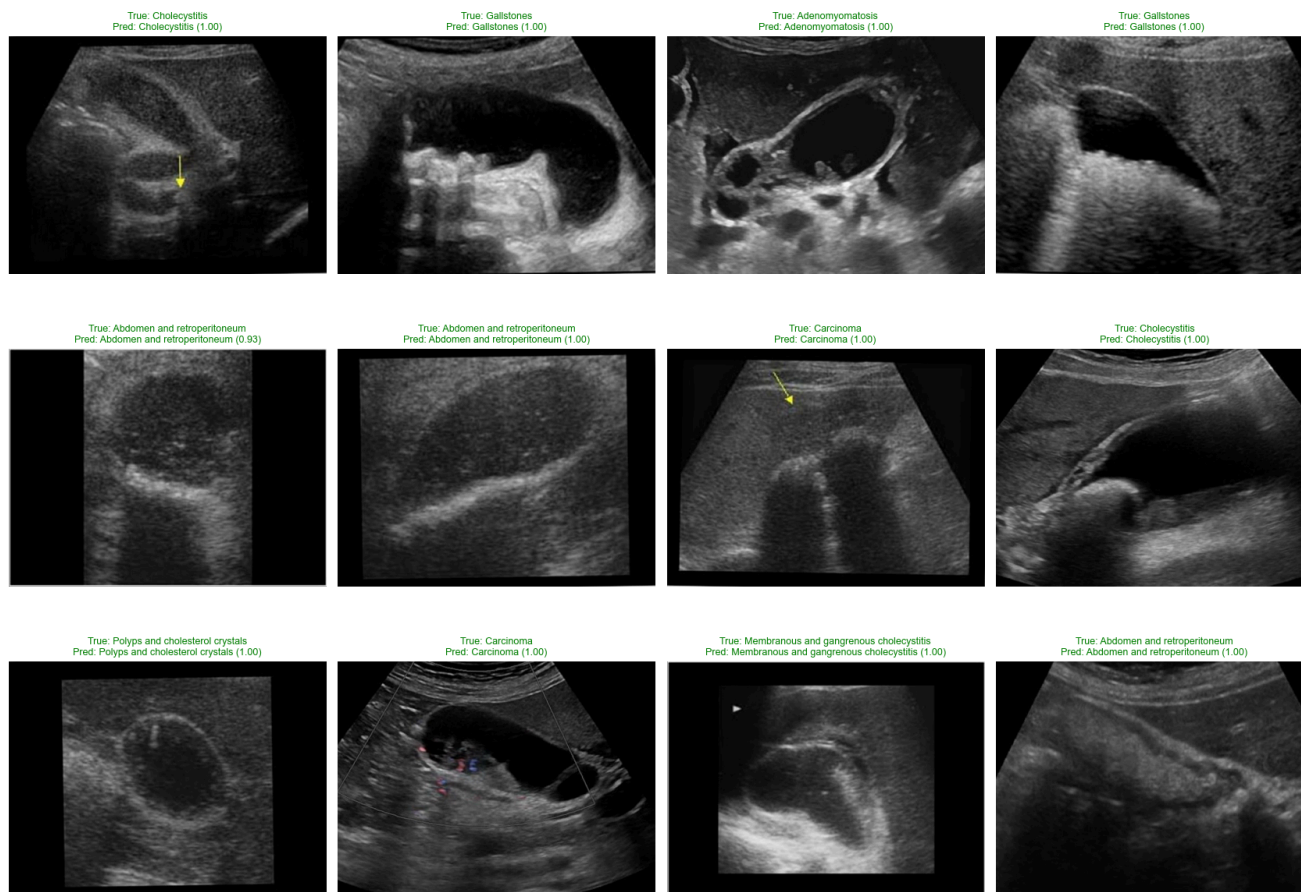


Figure 4. Sample prediction outputs

6. Conclusion

The Residual Attention CNN successfully classified gallbladder ultrasound images into 9 classes. It achieved 97.85% test accuracy and a 97.93% macro F1 score. The results show that residual connections and channel attention can learn useful features from this ultrasound image dataset.

7. GitHub Repo Link